JIARUI WANG

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Summary

My research focuses on memory architectures for large language models, aiming to decouple knowledge storage from reasoning capabilities and enable models to continually acquire, retain, and update knowledge. To this end, I develop scalable, modular parametric memories that can be trained independently and integrated with modern LLM backbones. These efforts involve pretraining, supervised fine-tuning, large-scale data pipelines, and inference systems. Most recently, I led the research and development of *Intern-S2-Preview*'s memory architecture for continual domain expansion.

Education

Shanghai Jiao Tong University, Shanghai, China — Sept. 2025 – Jun. 2030
Ph.D. in Computer Science and Technology
Advisor: Prof. **Zhouhan Lin** (LUMIA Group)

Shanghai Jiao Tong University, Shanghai, China — Sept. 2021 – Jun. 2025
B.S. in Computer Science and Technology

Publications

- **MemSFT: Mitigating Alignment Tax with an External Parametric Memory**
  Jiarui Wang[*], Xiang Shi[*], Jiaqi Cao, Rubin Wei, Xiquan Wang, Hao Sun, Jingzhi Wang, Zhiqi Yang, Qipeng Guo, Bowen Zhou, Zhouhan Lin
  arXiv:2607.25614, 2026.
- **Intern-S2-Preview: Scientific Agentic Foundation Model**
  Intern-S2-Preview Team, Shanghai AI Laboratory; Co-author; *arXiv:2608.13505, 2026.*
- **Memory Decoder at Scale: A Pretrained, Parametric Long-Term Memory**
  Rubin Wei, Jiaqi Cao, **Jiarui Wang**, Junming Zhang, Qipeng Guo, Bowen Zhou, Zhouhan Lin
  arXiv:2607.27919, 2026.
- **MLP Memory: A Retriever-Pretrained Memory for Large Language Models**
  Rubin Wei[*], Jiaqi Cao[*], **Jiarui Wang**, Jushi Kai, Qipeng Guo, Bowen Zhou, Zhouhan Lin
  ICLR 2026.
- **Memory Decoder: A Pretrained, Plug-and-Play Memory for Large Language Models**
  Jiaqi Cao[*], **Jiarui Wang**[*], Rubin Wei, Qipeng Guo, Kai Chen, Bowen Zhou, Zhouhan Lin
  NeurIPS 2025.

[*]Equal contribution.

Research Experience

LLM Research Intern, Shanghai AI Lab — Aug. 2025 – Present

- Led research and development of *Intern-S2-Preview*'s memory architecture, including model training and production deployment, to support continual domain expansion through independently trained, plug-and-play memories.
- Developed supporting infrastructure, including vLLM-based inference adaptation and optimization, large-scale data pipelines, and retrieval components for new LLM architectures.

Research Assistant, DDST Lab, Shanghai Jiao Tong University — May. 2023 – Nov. 2023

- Developed DebugPilot, an interactive debugging system that automates error diagnosis and debugging procedure generation using program analysis and probabilistic modeling.

Honors and Awards

Second Place, RoboCup China Open, Small Size League, 2024
Third Prize, Prototype System Competition, CCF ChinaSoft, 2023
First Prize (Provincial), Senior Division, National Olympiad in Informatics in Provinces (NOIP), 2018